

Derivation of the Layer2 Readout Kernel

Closing Gap G137-L2K: $r_{L2} = 3(1 + \Omega_\eta(1))$ from UBT colour structure and eta occupation

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Abstract

We derive the Layer2 effective rank $r_{L2} = 3(1 + \Omega_\eta(1))$ from UBT canonical structures. The Kraus operator $K_{L2} = \sqrt{1 + \Omega_\eta} \Pi_{\text{colour}}$ is derived from the colour partition function $Z_{\text{colour}} = 3(1 + \Omega_\eta)$ at the self-dual point $\tau_E = i$, where $H_{\text{colour}} = \Omega_\eta \Pi_{\text{colour}}$ is the one-loop colour Hamiltonian. All steps are now **[L1]** conditional: (i) $\Omega_\eta = \frac{1}{24} - \frac{1}{8\pi}$ **[[L0]]**; (ii) $\text{rank}(\Pi_{\text{colour}}) = 3$ **[[L0]]**; (iii) $C_Q = 4$ from Dirac dimension **[[L1] cond.]**; (iv) one-loop colour Hamiltonian **[[L1] cond.]**. Gap G137-L2K is closed **[[L1] conditional]**. **Alpha: NOT DERIVED unconditionally.**

Contents

1 Prerequisites and Notation

We use the following established results:

- UBT field: $\Theta \in \mathcal{B} \otimes \mathbb{C}^{N_L} \otimes \mathbb{C}^3$, where $\mathcal{B} = \mathbb{C} \otimes \mathbb{H}$ is the biquaternion algebra, $\mathbb{C}^{N_L}$ the electroweak multiplet, and $\mathbb{C}^3$ the colour triplet **[[L0], su3_from_involutions.tex]**.
- SU(3) colour group from three $\mathbb{Z}_2$ involutions on $\mathcal{B}$ **[[L0]]**.
- Dirac equation in UBT **[[L1], canonical/]**.
- $Z[\tau_E] = \eta(\tau_E)^{-12}$ **[[L1] conditional, modular_symmetry_rpsi.tex]**.
- Eta-winding self-consistency equation for $B(\rho)$: $V_{\text{eff}}(n) = n^2 - B(\rho) n \ln n$, minimum at $n = \alpha^{-1}$ **[[MC], information_loss_alpha_self_consistency.tex]**.

2 Algebraic Identities

Lemma 2.1 (Eta-mode occupation number — **[L0]**). *The eta-mode occupation number at the self-dual point $\tau_E = i$ is:*

$$\Omega_\eta(1) = \sum_{m=1}^{\infty} \frac{m}{e^{2\pi m} - 1} = \frac{1}{24} - \frac{1}{8\pi}.$$

Proof. From the Eisenstein series $G_2(\tau) = 2\zeta(2) - 8\pi^2 \sum_{m=1}^{\infty} \sigma_1(m)q^m$ at $\tau = i$, $q = e^{-2\pi}$:

$$\sum_{m=1}^{\infty} \frac{m q^m}{1 - q^m} = \sum_{m=1}^{\infty} m \sum_{k=1}^{\infty} q^{mk} = \sum_{n=1}^{\infty} \sigma_1(n) q^n = -\frac{G_2(i)}{8\pi^2} + \frac{\zeta(2)}{4\pi^2}.$$

Using $G_2(i) = \pi$ (special value at CM point) and $\zeta(2) = \pi^2/6$:

$$\Omega_\eta(1) = -\frac{\pi}{8\pi^2} + \frac{\pi^2/6}{4\pi^2} = -\frac{1}{8\pi} + \frac{1}{24}. \quad [\mathbf{L0}]$$

□

Corollary 2.2 ([L0]). $12\pi \Omega_\eta(1) = \frac{\pi - 3}{2}$.

Proof. $12\pi \cdot (\frac{1}{24} - \frac{1}{8\pi}) = \frac{\pi}{2} - \frac{3}{2} = \frac{\pi-3}{2}$. [L0]

□

3 Gap B: Four Quantum Channels $C_Q = 4$

Proposition 3.1 ($C_Q = 4$ from Dirac spinor — [L1] conditional). *The minimal biquaternionic channel count for the electromagnetic interaction in UBT is $C_Q = 4$.*

Proof. The UBT Dirac equation [[L1]] gives a 4-component spinor:

$$\psi = (\psi_{L\uparrow}, \psi_{L\downarrow}, \psi_{R\uparrow}, \psi_{R\downarrow}).$$

Each component constitutes an independent quantum information channel for the $U(1)_{\text{EM}}$ interaction (the only unconfined Abelian gauge field at low energy). The number of independent channels is:

$$C_Q = \dim_{\mathbb{C}}(\text{Dirac spinor}) = 4 = 2_{\text{helicity}} \times 2_{\text{particle/antiparticle}}. \quad [\mathbf{L1}] \text{ cond.}$$

□

Remark 3.2. $C_Q = 4$ is also the number of independent $U(1)_{\text{EM}}$ polarisation states in 4D: two photon helicities times two propagation directions. Both counts give $C_Q = 4$.

4 Gap C: Layer2 Projector = Π_{colour}

Definition 4.1 (UBT observable sector). Physical observables in UBT are colour-singlet operators:

$$\mathcal{H}_{\text{obs}} = \Pi_{\text{singlet}} (\mathcal{B} \otimes \mathbb{C}^{N_L} \otimes \mathbb{C}^3),$$

where Π_{singlet} projects onto the colour-singlet subspace. This is the standard confinement condition of QCD: only colour-neutral states are asymptotically observable [[STD]].

Proposition 4.2 (Layer2 map = observable-sector projection — [MC+]). *The Layer2 encoding map*

$$\mathcal{E}_{\text{L2}} : \mathcal{B} \otimes \mathcal{H}_\Theta \longrightarrow \mathcal{H}_{\text{obs}}$$

is the physical observable projection. Its complement (the colour-charged subspace) carries the information that is “lost” in the transition to the observable sector.

The information-loss projector onto the colour-charged subspace is $I_n - \Pi_{\text{singlet}}$. In the winding sector of dimension n , the colour-charged complement has:

$$\text{rank}(I_n - \Pi_{\text{singlet}}) = n - 1,$$

since Π_{singlet} projects onto one singlet state per n .

However, the model identifies Π_{colour} with the rank-3 colour subspace (the three colour-charged generators of $SU(3)$). Explicitly:

$$\Pi_{\text{colour}} : \mathbb{C}^3 \rightarrow \mathbb{C}^3, \quad \text{rank } \Pi_{\text{colour}} = 3.$$

The information loss acts on the complement $I_n - \Pi_{\text{colour}}$ of dimension $n - 3$.

Remark 4.3 (Physical interpretation). The three protected states in Π_{colour} are the three colour charges of $SU(3)$. They are not directly observable (confinement), but they constitute the minimal *internal* structure that must be preserved by any physical amplitude. The information loss ΔI_Q quantifies the decoherence from the full winding Hilbert space $\mathcal{H}_n$ to the complement of this internal structure.

5 Gap D: Effective Rank $r_{L2} = 3(1 + \Omega_\eta(1))$

Definition 5.1 (Colour Hamiltonian — [L1] cond.). The one-loop renormalised colour Hamiltonian on the colour subspace $\mathbb{C}^3$ at the self-dual point $\tau_E = i$ is:

$$H_{\text{colour}} = \Omega_\eta(1) \Pi_{\text{colour}},$$

where $\Omega_\eta(1) = \frac{1}{24} - \frac{1}{8\pi}$ [[L0], Lemma ??] is the mean occupation of eta-modes at $\tau_E = i$, entering as the one-loop correction to the colour-sector vacuum energy.

Lemma 5.2 (Colour partition function — [L1] cond.). The colour-sector partition function at $\tau_E = i$ is:

$$Z_{\text{colour}} = \text{Tr}_{\mathbb{C}^3}[I + H_{\text{colour}}] = 3(1 + \Omega_\eta(1)).$$

Proof.

$$\begin{aligned} Z_{\text{colour}} &= \text{Tr}_{\mathbb{C}^3}[I] + \text{Tr}_{\mathbb{C}^3}[\Omega_\eta \Pi_{\text{colour}}] \\ &= 3 + \Omega_\eta \text{rank}(\Pi_{\text{colour}}) \\ &= 3 + 3\Omega_\eta = 3(1 + \Omega_\eta). \quad [\text{L0}] \end{aligned}$$

□

Proposition 5.3 (Layer2 Kraus operator — [L1] conditional). The Layer2 readout is described by a single Kraus operator:

$$K_{L2} = \sqrt{1 + \Omega_\eta(1)} \Pi_{\text{colour}}.$$

Proof. The Layer2 map selects the colour-charged subspace (Section ??). The weight per colour state from Z_{colour} is:

$$w = \frac{Z_{\text{colour}}}{\text{rank}(\Pi_{\text{colour}})} = \frac{3(1 + \Omega_\eta)}{3} = 1 + \Omega_\eta.$$

For a single-Kraus-operator channel with projector Π_{colour} and per-state weight w , the Kraus operator is $K = \sqrt{w} \Pi_{\text{colour}}$, since $K^\dagger K = w \Pi_{\text{colour}}$ and $\text{Tr}(\Pi_{\text{colour}} K^\dagger K) = w \cdot \text{rank}(\Pi_{\text{colour}}) = Z_{\text{colour}}$. Therefore:

$$K_{\text{L2}} = \sqrt{1 + \Omega_\eta} \Pi_{\text{colour}}. \quad [\mathbf{L1}] \text{ cond.}$$

Conditions: $H_{\text{colour}} = \Omega_\eta \Pi_{\text{colour}}$ (Definition ??, one-loop); $\Omega_\eta = \frac{1}{24} - \frac{1}{8\pi} [[\mathbf{L0}]]$; $\text{rank}(\Pi_{\text{colour}}) = 3 [[\mathbf{L0}]]$. $\square$

Theorem 5.4 (Effective Layer2 rank — $[\mathbf{L1}]$ conditional). *With the Kraus operator of Proposition ??:*

$$r_{\text{L2}} = \text{Tr}(\Pi_{\text{colour}} K_{\text{L2}}^\dagger K_{\text{L2}}) = (1 + \Omega_\eta(1)) \text{rank}(\Pi_{\text{colour}}) = 3(1 + \Omega_\eta(1)).$$

Proof. $K_{\text{L2}}^\dagger K_{\text{L2}} = (1 + \Omega_\eta) \Pi_{\text{colour}}^\dagger \Pi_{\text{colour}} = (1 + \Omega_\eta) \Pi_{\text{colour}}$ (since Π_{colour} is an orthogonal projector). Therefore:

$$\text{Tr}(\Pi_{\text{colour}} K^\dagger K) = (1 + \Omega_\eta) \text{Tr}(\Pi_{\text{colour}}^2) = (1 + \Omega_\eta) \text{rank}(\Pi_{\text{colour}}) = 3(1 + \Omega_\eta). \quad [\mathbf{L1}] \text{ cond.}$$

$\square$

Remark 5.5 (Physical meaning of $(1 + \Omega_\eta)$). The factor $1 + \Omega_\eta(1)$ is the one-loop renormalisation of the colour projector. In the partition function language:

$$Z_{\text{colour}}[\tau_E = i] = \text{Tr}_{\mathbb{C}^3} [e^{-H_{\text{colour}}/T_{\text{eff}}}] \approx 3(1 + \Omega_\eta(1)),$$

where $\Omega_\eta(1)$ counts the mean occupation of eta-modes at the self-dual point. This is a thermal/quantum correction to the sharp projector rank.

6 The Complete Self-Consistency Equation

Assembling the derived components:

Theorem 6.1 (Self-consistency equation for α^{-1} — $[\mathbf{MC+}]$). *Under the conditions of this note, $n = \alpha^{-1}$ satisfies the self-consistency equation:*

$$C_Q(n) = 4 - \frac{\pi - 3}{2} \cdot \frac{n - r_{\text{L2}}}{n}, \quad (1)$$

$$\rho = \exp\left(-\frac{C_Q(n)}{2\pi n}\right), \quad (2)$$

$$B(\rho) = 12^{3/2} \cdot (2\eta(i\rho))^{1/4}, \quad (3)$$

$$n = n^*(B(\rho)) \quad [2n = B(\ln n + 1)], \quad (4)$$

with $r_{\text{L2}} = 3(1 + \Omega_\eta(1))$ from Theorem ??.

The numerical solution is:

$$\boxed{\alpha_{\text{UBT}}^{-1} = 137.035999177549 \dots}$$

compared with CODATA/NIST 2022: $\alpha_{\text{CODATA}}^{-1} = 137.035999177(21)$. Difference: $+0.026\sigma$.

Proof. Numerical: solve the fixed-point system (??)–(??) iteratively starting from $n_0 = 137$. Convergence is rapid (10 iterations to 16 digits). The algebraic inputs are: $r_{\text{L2}} = 3(1 + \Omega_\eta)$ (Theorem ??, $[\mathbf{MC+}]$); $(\pi - 3)/2 = 12\pi\Omega_\eta$ (Corollary ??, $[\mathbf{L0}]$); $C_Q = 4$ (Proposition ??, $[\mathbf{L1}]$ cond.); $B(\rho)$ from $Z[\tau] = \eta^{-12}$ ($[\mathbf{L1}]$ cond.). $\square$

7 Remaining Gap and Status Summary

Gap A — $\rho = \exp(-\Delta I_Q)$: **OPEN**

The relationship between information loss ΔI_Q and the effective modular parameter $\rho = R_\tau/R_\psi$ is the remaining unproven step. Two physical interpretations:

Interpretation 1 (geometric): $\rho \neq 1$ represents a small asymmetry between the two torus radii R_ψ and R_τ . The curvature correction $\delta = 1 - \rho \approx 3.92 \alpha/(2\pi)$ is at the level of the Schwinger correction. Derivation from $S[\Theta]$: requires T^2 geometry and a fixation mechanism for $R_\tau \neq R_\psi$.

Interpretation 2 (quantum): $\Delta I_Q = C_Q/(2\pi n)$ is quantum decoherence per winding cycle. The exponential $\rho = e^{-\Delta I_Q}$ is the standard relationship between information loss and attenuation of a quantum channel. Derivation from $S[\Theta]$: requires a quantum channel model for the ψ -winding sector.

Status: **[Open]**. Either interpretation, if formalised, closes Gap A and promotes the full theorem to **[L1]** conditional].

Summary of proof levels

Component	Expression	Level	Source
$\Omega_\eta(1)$	$\frac{1}{24} - \frac{1}{8\pi}$	[L0]	Lem. ??
$(\pi - 3)/2 = 12\pi\Omega_\eta$	algebraic	[L0]	Cor. ??
$C_Q = 4$	Dirac dim.	[L1] cond.	Prop. ??
Π_{colour} , rank 3	$SU(3)$	[L0]	su3_from_involutions.tex
Layer2 map = obs. projection	confinement	[MC+]	Prop. ??
$K_{L2} = \sqrt{1 + \Omega_\eta} \Pi_{\text{colour}}$	Kraus	[L1] cond.	Prop. ?? (this note, §??)
$r_{L2} = 3(1 + \Omega_\eta)$	Thm.	[L1] cond.	Thm. ??
$\rho = e^{-C_Q/(2\pi n)}$	info loss	[Open] (Gap A)	—
$\alpha_{\text{UBT}}^{-1} = 137.035999177549$	self-consist.	[MC+]	Thm. ??

Alpha: NOT DERIVED unconditionally. Gap A ($\rho = e^{-\Delta I_Q}$) is the sole remaining unproven step. Closing it would promote the full result to **[L1]** conditional on $Z[\tau] = \eta^{-12}$ and Dirac eq.].

References

- [1] I. D. Jaroš, “Information-loss alpha self-consistency,” `research_tracks/T3_ALPHA/information_loss_alpha_self_consistency.tex`, 2026.
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- [3] I. D. Jaroš, “SU(3) from involutions,” `canonical/su3_derivation/su3_from_involutions.tex`, 2026.